

WASP-43b

R-2678

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_130415 · created 2026-08-30T23:19:52+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456397.65797 +/- 0.00034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1822 +/- 0.0076
Transit depth (Rp/Rs)^2	3.32 +/- 0.28 [%]
Orbital Inclination (inc)	83.79 +/- 0.84
Airmass coefficient 1 (a1)	1.598 +/- 0.014
Airmass coefficient 2 (a2)	-0.342 +/- 0.054
Scatter in the residuals of the lightcurve fit is	0.86 %
Best Comparison Star	None
Optimal Aperture	3.26
Optimal Annulus	5.96
Transit Duration (day)	0.05 +/- 0.001

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1822 $\pm$ 0.0076 vs 0.1594 (deviation 2.87 σ)
Duration fit vs published	0.05 d vs 0.0512 d (deviation 1.2 σ)
Mid-time O-C	-1.5 min
Depth SNR	22.9
Residual scatter	0.86 %

Light curve

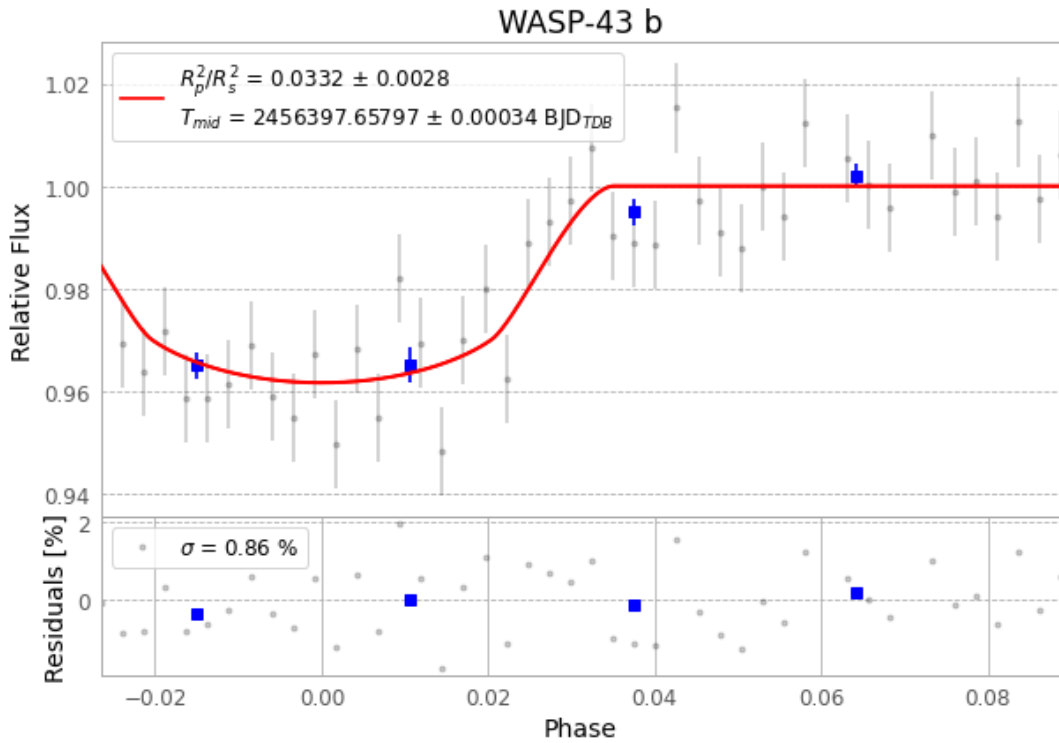


Figure 1 — FinalLightCurve_WASP-43 b_2013-04-14.png (SHA-256 a27cb78b57c2f82b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [326, 301], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f6c3e7693f4fcf58...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

46 FITS frames (30 MB), WASP-43130415030912.FITS → WASP-43130415052413.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-130415.log (ba4ee362f3e8...), FinalLightCurve_WASP-43 b_2013-04-14.png (a27cb78b57c2...), FinalLightCurve_WASP-43 b_2013-04-14.csv (aaf376130119...)

Compute resources

Wall time 297.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 23:19Z.